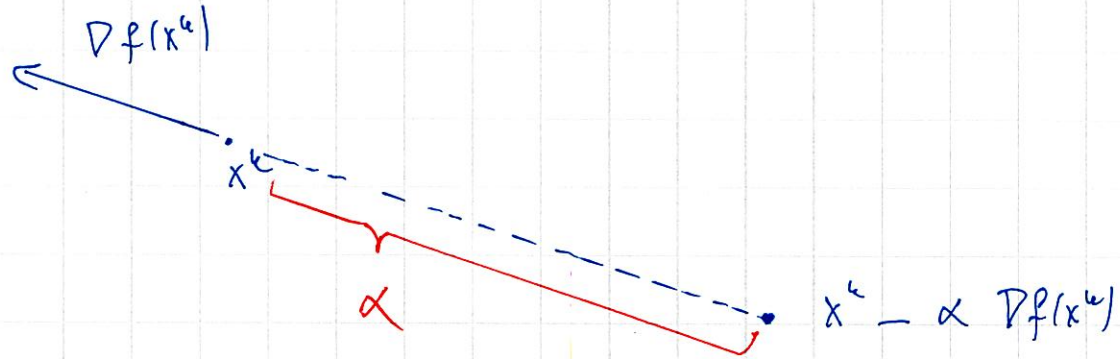
Using Single-Dimensional Optimization to find Best step size During Gradient Descent

- We are minimizing $f: \mathbb{R}^n \to \mathbb{R}$ by using gradient descent. At iteration k of gradient descent we are at point x^k.
- We compute $\nabla f(x^k)$ and we want to move against the gradient for a certain step size α



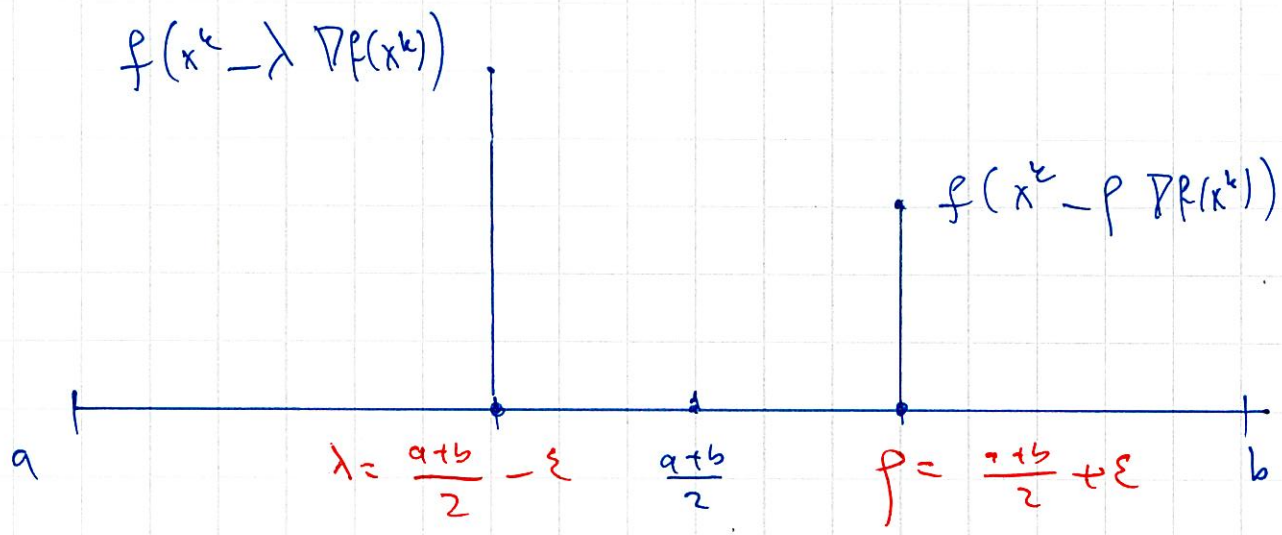
- Choose the step size α such that the value of the function at the next point we reach is as small as possible

trying to find

$$\min_{\alpha \geq 0} f(\underbrace{x^k}_{\text{known}} - \alpha \underbrace{\nabla f(x^k)}_{\text{known}})$$

- This is a single dim. optimization problem.

- Say we are at iteration k of gradient descent and we are at point x^k (known)
- Compute $\nabla f(x^k)$ (known)
- Let's use single-dimensional optimization to find best step size
- Think about what bisection search would look like to find the best step size



- Minimize $f(x) = 4 \ln(e^{2x_1} + e^{4x_2}) - 6x_1 - 4x_2$ using gradient descent

$$\nabla f(x) = \left(\frac{8 e^{2x_1}}{e^{2x_1} + e^{4x_2}} - 6 , \frac{16 e^{4x_2}}{e^{2x_1} + e^{4x_2}} - 4 \right)$$

Use bisection search to find the best step size at each iteration.

Constrained optimization:

- Consider a constrained optimization problem of the form:

$$\min \quad f(x)$$
$$\text{st}$$
$$g_1(x) \leq 0$$
$$\vdots$$
$$g_n(x) \leq 0$$
$$h_1(x) = 0$$
$$\vdots$$
$$h_m(x) = 0,$$

where $f, g_1, \dots, g_n, h_1, \dots, h_m : \mathbb{R}^n \to \mathbb{R}$

- In all the crude constrained optimization methods we will consider, we will convert the constrained optimization problem into an equivalent unconstrained optimization problem and use our standard gradient descent algorithm for unconstrained optimization.

Penalty Function Method:

- Moves the constraints into the objective function by hugely penalizing the violations of the constraints.

- $\min \; f(x)$

  st $\; g_1(x) \leq 0$

  $\vdots$

  $g_n(x) \leq 0$

  $h_1(x) = 0$

  $\vdots$

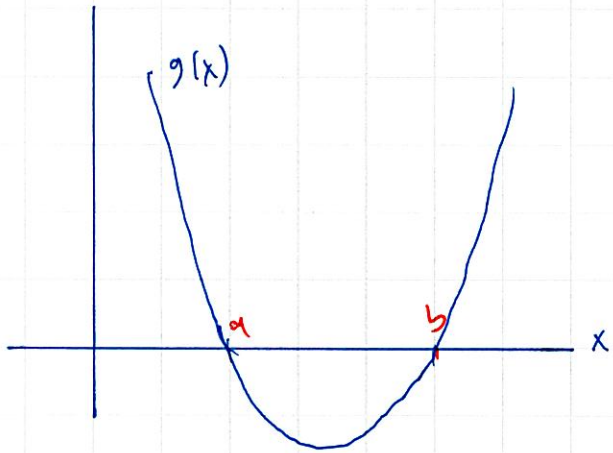
  $h_m(x) = 0$

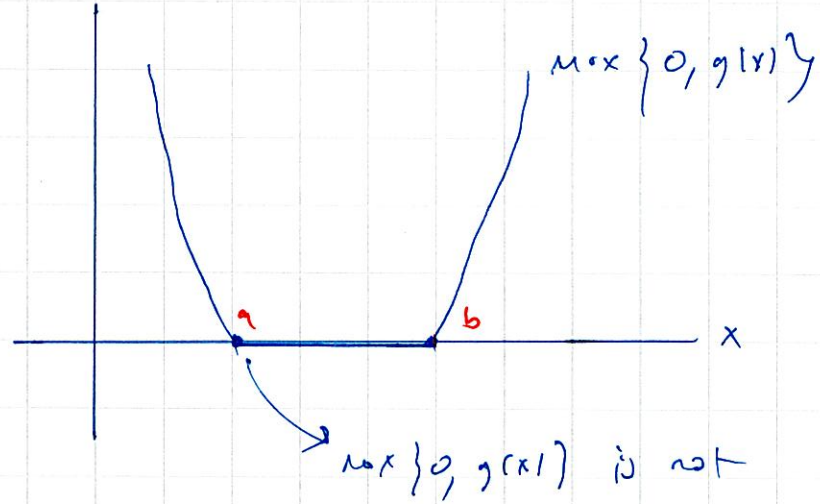
- Solve the unconstrained problem

$$\min \; f(x) + \sum_{i=1}^{n} \theta_i \left(\max\{0, g_i(x)\}\right)^2 + \sum_{i=1}^{m} \gamma_i \left(h_i(x)\right)^2,$$

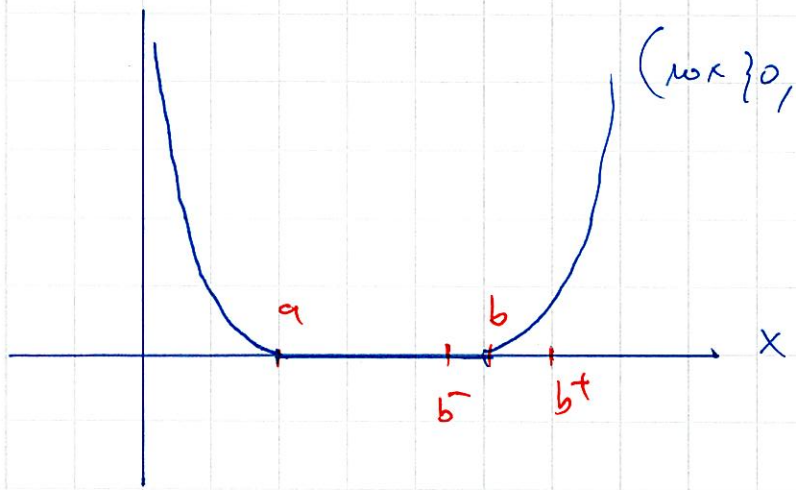
where $\theta_1, \ldots, \theta_n, \gamma_1, \ldots, \gamma_m$ are large penalty parameters

- Why did we square the penalty term of the inequality constraint?





max{0, g(x)} is not differentiable



$(\max\{0, g(x)\})^2 = r(x)$

$r'(b^-) = 0$

$r(x)$ to the right of b behaves like $g^2(x)$

$r'(x)$ to the right of b behaves like $2g(x) \cdot g'(x)$

$r'(b^+) = 2g(b^+) \cdot g'(b^+) = 0$ when b^+ is very close to b

1. Choose the initial penalty parameters $\theta_1^1, \dots, \theta_n^1, \gamma_1^1, \dots, \gamma_m^1$, tolerance parameter ε, magnification parameter $\beta > 1$, set $k=1$

2. Solve the problem

$$\min_x \left\{ f(x) + \sum_{i=1}^{n} \theta_i^k \left(\max\{0, g_i(x)\}\right)^2 + \sum_{i=1}^{m} \gamma_i^k \left(h_i(x)\right)^2 \right\}$$

Let x^k be the optimal solution

3. If $\sum_{i=1}^{n} \theta_i^k \left(\max\{0, g_i(x^k)\}\right)^2 + \sum_{i=1}^{m} \gamma_i^k \left(h_i(x^k)\right)^2 \leq \varepsilon$, then stop

else

set $\theta_i^{k+1} = \beta\, \theta_i^k$ $i=1,\dots,n$ $\gamma_i^{k+1} = \beta\, \gamma_i^k$ $i=1,\dots,m$ and go to step 2.

Example:

$$\min \quad (x_1 - 2)^4 + (x_1 - 2x_2)^2$$
$$\text{st} \quad x_1^2 - x_2 \leq 0$$